

Matrices and Determinants

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<https://www.alphaclasses.com/>

Educational

<https://github.com/rajatkalia1982>
<https://sourceforge.net/u/rajatkalia/profile>
<https://www.youtube.com/@rajatkaliaeducation>

Store

<https://www.lulu.com/spotlight/rajatkalia11>
<https://amazon.com/author/rajatkalia>

Sports

<https://www.youtube.com/@indiangamersorganization>
<https://www.facebook.com/groups/igohn>
<https://india.voobly.com/>

Social

<https://www.goodreads.com/sardarbhagatsingh>
<https://x.com/RajatKalia1>
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<https://www.researchgate.net/profile/Rajat-Kalia>
<https://www.w3profile.com/rajatkalia/>
<https://www.reddit.com/user/rajatkalia/>

Blogs

<https://rk-e.blogspot.in/> [Physics]
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<https://battalion-7.blogspot.com/> [Armed Forces]
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<https://tmm18.blogspot.com/> [Meditation]

Freelancing

<https://www.guru.com/freelancers/rajat-kalia>
<https://www.codecademy.com/profiles/rajatkalia>
<https://leetcode.com/u/rajatkalia1/>
<https://www.scribd.com/user/308680256/Rajat-Kalia>

Part I

Matrices Problem Set

Chapter 1

JEE Mains Problems

Q1 : Let $A = \begin{bmatrix} 1 & 2 \\ 1 & \alpha \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 3 \\ \beta & 2 \end{bmatrix}$. If $A^2 - 4A + I = 0$ and $B^2 - 5B - 6I = 0$, then among the two statements :

$$(S1) : [(B - A)(B + A)]^T = \begin{bmatrix} 13 & 15 \\ 7 & 10 \end{bmatrix}$$

$$(S2) : \det(\operatorname{adj}(A + B)) = -5,$$

[JEE Mains 2nd Apr 2026 Shift 1]

(A) Only (S1) is correct

(B) Only (S2) is correct

(C) Both (S1) and (S2) are correct

(D) Both (S1) and (S2) are wrong

{Solution : <https://www.youtube.com/watch?v=uTUbsovqU8>}